

Renin-Aldosterone Profiles and Cardiorenal Outcomes in Hypertension

A Nationwide Cohort Study

Christine Ljungberg, MD,^a Björn Carlsson, MD, PhD,^b Poyan Shojaiyan, MD,^c Stefan Franzén, PhD,^d
Jesper N. Bech, MD, PhD,^e Ken Lee Chin, PhD,^f Christian F. Christiansen, MD, PhD,^a Reimar W. Thomsen, MD, PhD^a

ABSTRACT

BACKGROUND Renin-independent aldosterone production may contribute to adverse cardiorenal outcomes, but population-based evidence remains limited.

OBJECTIVES In this study, we sought to investigate the association of renin-aldosterone profiles with cardiorenal events in adults with hypertension.

METHODS This cohort study used linked Danish health registries with complete laboratory test results from January 2017 to November 2024. All adults with hypertension selected by clinicians for aldosterone and renin testing in the study period were included. Endpoints included kidney function decline and cardiovascular outcomes. We used Cox regression and restricted cubic spline models adjusted for age, sex, index year, comorbidities, and comedications.

RESULTS Among 12,650 adults with hypertension, 1,840 (15%) had an aldosterone-to-renin ratio (ARR) of 27.7 to <70 pmol/mIU, 480 (4%) 70 to <138.7 pmol/mIU, and 310 (2%) ≥138.7 pmol/mIU. Median follow-up was 3.6 years. Higher ARR, higher aldosterone, and lower renin levels were associated with increased adjusted HRs (aHRs) of rapid kidney function decline (eGFR decrease of at least 5 mL/min/1.73 m² per year), kidney failure or ≥40% eGFR decline, and major adverse cardiovascular events (MACE) in cubic spline models. Risk of kidney function decline outcomes varied across combinations of aldosterone and renin levels, with higher risk observed in individuals with elevated aldosterone in the context of low renin. Analyses using predefined ARR levels showed similar trends. Specifically, the aHRs for rapid kidney function decline were 1.27 (95% CI: 1.17-1.38), 1.98 (95% CI: 1.75-2.25), and 2.66 (95% CI: 2.31-3.08) for ARR 27.7 to 70.0, 70.0 to 138.7, and >138.7 pmol/mIU, respectively. For kidney failure or ≥40% eGFR decline and for MACE, the aHRs were not elevated at ARR 27.7-70.0 pmol/mIU, but increased at higher ARR levels.

CONCLUSIONS Adults with hypertension and renin-independent aldosterone production are at increased risk of rapid kidney function decline as well as, at higher ARR levels, kidney failure and MACE. Although clinical practice guidelines recommend distinct thresholds for diagnosing primary aldosteronism, our findings underscore that the risk for kidney and cardiovascular outcomes in adults with hypertension manifests as a continuous gradient that parallels the magnitude of renin-independent aldosterone production. (JACC. 2026;■:■-■) © 2026 by the American College of Cardiology Foundation.

From ^aAarhus University and the Aarhus University Hospital, Department of Clinical Epidemiology, Aarhus, Denmark;

^bCardiovascular, Renal, and Metabolism, Translational Science and Early Clinical Development, AstraZeneca, Gothenburg, Sweden; ^cMedical Evidence, AstraZeneca, Stockholm, Sweden; ^dBioPharmaceuticals Medical Evidence Statistics, AstraZeneca, Gothenburg, Sweden; ^eUniversity Clinic in Nephrology and Hypertension, Department of Medicine, Godstrup Hospital and Aarhus University, Aarhus, Denmark; and ^fBiopharmaceuticals Medical Cardiovascular, Renal, and Metabolism Evidence Strategy, AstraZeneca, Gothenburg, Sweden.

The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

Manuscript received August 22, 2025; revised manuscript received January 15, 2026, accepted January 20, 2026.

**ABBREVIATIONS
AND ACRONYMS****ARR** = aldosterone-to-renin ratio**MACE** = major adverse cardiovascular events

Aldosterone, a mineralocorticoid hormone produced by the adrenal glands, plays a central role in maintaining kidney and cardiovascular homeostasis by regulating sodium retention, potassium excretion, and blood pressure.¹

However, beyond its physiologic effects, inappropriate or excessive aldosterone activity, independently from renin regulation, has been implicated in a spectrum of renal and cardiovascular pathologies.²⁻⁶ Although primary aldosteronism represents the most well characterized form of renin-independent aldosteronism, growing evidence suggests that renin-independent aldosterone production extends beyond primary aldosteronism and may be a more prevalent, yet underrecognized, driver of adverse outcomes.⁷⁻¹¹

Recent studies have challenged the traditional view of primary aldosteronism as a binary condition, instead supporting a broader continuum of inappropriate aldosterone activity.⁷⁻¹¹ Even in individuals without hypertension or hypokalemia, suppressed renin and elevated aldosterone levels have been associated with adverse renal and cardiac remodeling, arterial stiffness, and increased renal and cardiovascular risk.^{9,11-14} However, the clinical importance of this broader spectrum remains incompletely understood as current screening approaches primarily focus on high risk populations, potentially overlooking many affected individuals.

The Framingham Offspring Study (FOS), Jackson Heart Study, Chongqing Primary Aldosteronism Study (COMPASS), CARTaGENE, Multi-Ethnic Study of Atherosclerosis (MESA), and Atherosclerosis Risk in Communities (ARIC) study have all reported associations between renin-independent aldosterone production and increased cardiovascular risk, though findings vary depending on population characteristics and aldosterone and renin thresholds used.^{6,7,15-22} Of note, population-based data on the epidemiology of renin-aldosterone profiles in routine clinical care remain limited. Targeted treatment strategies such as mineralocorticoid receptor antagonists and aldosterone synthase inhibitors hold promise for mitigating aldosterone-mediated organ damage, but their optimal use requires a clearer understanding of how renin-independent aldosterone production is identified and of its long-term consequences.^{9,23} Therefore, we aimed to investigate the impact of renin-independent aldosterone production on progression of kidney disease, cardiovascular disease, and all-cause mortality with the use of nationwide

CLINICAL PERSPECTIVE

- Renin-aldosterone physiology represents a continuous risk gradient, not a binary abnormality, transcending arbitrary thresholds.
- Kidney outcomes appear earlier and more consistently than cardiovascular outcomes.
- Even below classical thresholds for primary aldosteronism, there is a continuous risk gradient that parallels the magnitude of renin-independent aldosterone production.
- ARR-tested patients represent a high-risk clinical subset; results should be interpreted in this context.

population-based Danish health care registries with complete laboratory data.

METHODS

DESIGN AND DATA SOURCES. The Aldosterone Dysregulation Epidemiology Landscape (ADEL) study was designed to use Denmark's nationwide health care registries covering complete laboratory results for the entire population to examine the occurrence of aldosterone and renin testing, estimate the occurrence of renin-independent aldosterone production among the tested, and describe the clinical characteristics of tested adults and individuals with renin-independent aldosterone production and long-term rates of cardiorenal events and mortality among individuals undergoing aldosterone and renin testing. A comprehensive overview of occurrence and clinical characteristics has been reported elsewhere.²⁴

Denmark's tax-funded health care system ensures that data are systematically recorded across nationwide databases. Exact individual-level data linkage across all registries is possible through a unique personal registration number assigned to all Danish citizens at birth or upon immigration.^{25,26} Data were obtained from 4 national registries: the Danish Civil Registration System, which records data on residence, migration, and vital status of all Danish residents since 1968^{25,26}; the Danish National Patient Registry, which contains detailed data on all inpatient and outpatient hospital diagnoses and treatments since 1977²⁷; the Danish National Prescription Registry, which includes individual-level data on all

reimbursed prescriptions at Danish pharmacies since 1995²⁸; and the Registry of Laboratory Results for Research, which has recorded blood sample biochemistry data since 2015.²⁹

The study was registered at the Danish Data Protection Agency through registration at Aarhus University. According to Danish legislation, ethical approval is not required for strictly registry-based studies.

STUDY POPULATION. The study population consisted of Danish adults selected by treating clinicians for measurement of aldosterone and renin. As such, all findings reflect a clinically referred cohort, and the results should be interpreted in the context of this selective testing rather than a screened general population sample.

We first identified all direct aldosterone-to-renin ratio (ARR) measurements, all plasma-based aldosterone and renin measurements performed within a maximum of 7 days of each other in the entire Danish population from January 1, 2017, to October 31, 2024. The cohort was then restricted to adults (≥ 18 years) with hypertension and at least 1 year of continuous residency at the time of aldosterone and renin measurements. Hypertension was defined by a validated algorithm as either a prior hospital diagnosis of hypertension or pharmacologic treatment with prescriptions in at least 2 antihypertensive drug classes.³⁰ For each individual with hypertension, we selected the first aldosterone and renin test and assigned its date as the index date and, if a aldosterone-renin pair had been measured in an individual within 7 days, the test date was defined as the date of the last test in that window (Supplemental Figure 1).

EXPOSURE AND OUTCOME. The exposure of interest was renin-independent aldosterone production, assessed according to plasma aldosterone, renin, and aldosterone-to-renin ratio (ARR). Aldosterone, renin, and ARR were analyzed both as continuous and categorical variables. Given the uncertainties regarding the definition of primary aldosteronism and its clinical implications, we evaluated aldosterone, renin, and ARR levels using categorical dose-response strata. For ARR, we defined levels as <27.7 , 27.7 to <70.0 , 70.0 to <138.7 , and ≥ 138.7 pmol/mIU. Aldosterone concentrations were categorized as <139 , 139 to <277 , 277 to <554.8 , and ≥ 554.8 pmol/L. Renin concentrations were categorized as >15 , >8.2 to 15 , >5.0 to 8.2 , and ≤ 5.0 mIU/L. These levels were based on Endocrine Society criteria,³¹ the CONPASS and Framingham Offspring (FOS) studies,^{16,17} European Society of Hypertension guidelines,³² and the study by Townsend et al.^{33,34}

We assessed several renal and cardiac outcomes as well as mortality using nationwide hospital and laboratory registry data. The primary outcomes were rapid kidney function decline (an annual estimated glomerular filtration rate (eGFR) decrease of ≥ 5 mL/min/1.73 m² per year), and a Kidney Disease: Improving Global Outcomes-aligned composite outcome of end-stage kidney disease or a $\geq 40\%$ decline in eGFR.^{35,36} To minimize misclassification of acute kidney injury as chronic decline, only outpatient clinic eGFR measurements were included in the outcomes, excluding measurements from acute care settings such as inpatient admissions. Furthermore, for kidney function decline outcomes, 890 individuals without a baseline eGFR measurement were excluded. Other study outcomes included acute myocardial infarction, stroke, major adverse cardiovascular events (MACE) (a composite of myocardial infarction, stroke, and cardiovascular death), and all-cause mortality. Detailed variable definitions are provided in Supplemental Table 1.

STATISTICAL ANALYSES. We first described baseline characteristics in the study population stratified by ARR level. Follow-up began on the index date and continued until the occurrence of a clinical outcome or the end of the study, November 2024, whichever came first. For outcomes requiring longitudinal assessments such as rapid eGFR decline and 40% decline in eGFR, only individuals with both baseline and follow-up measurements were included in the analysis.

Incidence rates were computed for each outcome by dividing the number of events with the total person-time for different levels of ARR, aldosterone, and renin with 95% CIs derived using exact Poisson tests. We examined the association between renin-independent aldosterone production (including ARR, aldosterone, and renin) and the outcomes with the use of cause-specific Cox proportional hazard models, treating death as a censoring event. ARR, aldosterone, and renin were analyzed both continuously and as a categorical variable applying the predefined levels. In the models with continuous variable, the variable was included as a spline with 4 knots (placed at 5th, 35th, 65th, and 95th percentiles), and the results were visualized as estimated hazard ratios with 95% CIs. To further explore the joint association of aldosterone and renin, we used spline-based Cox models with interaction term to visualize the estimated hazard ratios across the distribution of aldosterone and renin in a 2-dimensional heatmap. The model was evaluated based on the Akaike information criterion.

TABLE 1 Patient Characteristics According to ARR Levels

	All ARR Tested (N = 12,650)	ARR <27.7 pmol/mIU (n = 10,020)	ARR ≥27.7 to <70 pmol/mIU (n = 1,840)	ARR ≥70 to <138.7 pmol/mIU (n = 480)	ARR ≥138.7 pmol/mIU (n = 310)
Age, y	58.2 (46.0-69.8)	58.3 (45.7-70.0)	58.3 (46.7-69.6)	57.9 (46.2-67.2)	57.9 (47.3-68.6)
Male	5,930 (47)	4,760 (48)	740 (40)	250 (52)	180 (58)
Index year					
2017–2018	3,400 (27)	2,760 (28)	460 (25)	110 (23)	70 (23)
2019–2020	3,290 (26)	2,580 (26)	480 (26)	130 (27)	100 (32)
2021–2022	3,380 (27)	2,630 (26)	540 (29)	130 (27)	80 (26)
2023–2024	2,580 (20)	2,040 (20)	370 (20)	100 (21)	70 (23)
Aldosterone dysregulation testing					
Aldosterone, pmol/L	176.0 (103.0-367.0)	144.0 (103.0-308.0)	259.7 (103.0-460.0)	451.0 (316.0-673.0)	769.0 (523.0-1109)
Renin, mIU/L	20.0 (8.5-50.0)	28.0 (13.0-65.0)	5.9 (2.9-11.0)	4.7 (3.2-7.0)	3.0 (1.9-5.0)
ARR, pmol/mIU	10.0 (4.1-22.9)	7.1 (3.2-13.4)	41.8 (33.2-54.2)	94.6 (80.2-112.8)	211.7 (169.4-300.5)
Hypertension related characteristics					
Hospital diagnosed hypertension	9,940 (79)	7,840 (78)	1,480 (80)	380 (79)	240 (77)
Duration of hypertension	5.7 (0.3-13.6)	5.2 (0.2-13.2)	6.0 (0.3-14.0)	8.4 (1.4-14.6)	10.9 (2.9-17.6)
Number of hypertensive drug classes used within the last year					
0	1,410 (11)	1,140 (11)	210 (11)	40 (8)	30 (10)
1	1,980 (16)	1,600 (16)	290 (16)	60 (13)	30 (10)
2	4,050 (32)	3,330 (33)	520 (28)	110 (23)	90 (29)
3	3,410 (27)	2,640 (26)	510 (28)	160 (33)	90 (29)
4–5	1,800 (14)	1,320 (13)	320 (17)	100 (21)	70 (23)
Comorbidities					
Ischemic heart disease	1,330 (11)	1,070 (11)	180 (10)	50 (10)	30 (10)
Cerebrovascular disease	1,270 (10)	1,020 (10)	160 (9)	50 (10)	40 (13)
Peripheral vascular disease	620 (5)	540 (5)	60 (3)	10 (2)	10 (3)
Atherosclerotic disease	2,730 (22)	2,200 (22)	360 (20)	90 (19)	70 (23)
Heart failure	550 (4)	460 (5)	60 (3)	20 (4)	10 (3)
Type 2 diabetes	1,980 (16)	1,600 (16)	260 (14)	70 (15)	50 (16)
Prediabetes	950 (8)	760 (8)	130 (7)	30 (6)	30 (10)
Dyslipidemia	4,820 (38)	3,890 (39)	640 (35)	170 (35)	120 (39)
Obesity	1,310 (10)	1,050 (10)	180 (10)	50 (10)	20 (6)
CKD	1,970 (16)	1,610 (16)	250 (14)	60 (13)	50 (16)
Adrenal gland disorder	860 (7)	650 (6)	120 (7)	50 (10)	40 (13)
Obstructive sleep apnea	760 (6)	590 (6)	110 (6)	40 (8)	30 (10)

Continued on the next page

The Cox models were adjusted for sex, age, index year, dyslipidemia, obesity, chronic obstructive pulmonary disease, type 2 diabetes, chronic kidney disease, metabolic-associated fatty liver disease, and baseline use of a range of individual medications including angiotensin-converting enzyme inhibitors, angiotensin II receptor blockers, angiotensin receptor-neprilysin inhibitor, calcium channel blockers, thiazide diuretics, loop diuretics, potassium-sparing diuretics, mineralocorticoid receptor antagonists, beta-blockers, alpha-blockers, metformin, sodium-glucose cotransporter 2 inhibitors, glucagon-like peptide 1 receptor agonists, statins, antiplatelets, and antithrombotic agents. The proportional hazards assumption was assessed with the use of Schoenfeld residuals, with *P* values of

proportional hazard assumptions reported in [Supplemental Table 2](#).

To protect patient privacy, as required by the Danish regulation, all numbers were rounded to the nearest 10 to avoid the possibility of back-calculating small numbers of individuals in subgroups, and incidence rates were calculated based on these rounded values.

ADDITIONAL ANALYSES. Several additional analyses were performed to assess the robustness of the findings. First, stratified analyses were conducted based on key baseline characteristics, including atherosclerotic cardiovascular disease, heart failure, and chronic kidney disease; potassium level (<3.5 and ≥3.5 mmol/L); and number of hypertensive

TABLE 1 Continued

	All ARR Tested (N = 12,650)	ARR <27.7 pmol/mIU (n = 10,020)	ARR ≥27.7 to <70 pmol/mIU (n = 1,840)	ARR ≥70 to <138.7 pmol/mIU (n = 480)	ARR ≥138.7 pmol/mIU (n = 310)
Medications					
ACE inhibitors	3,590 (28)	2,900 (29)	480 (26)	130 (27)	80 (26)
ARBs	6,010 (48)	4,740 (47)	850 (46)	250 (52)	160 (52)
ARNi	20 (0)	10 (0)	0 (0)	0 (0)	0 (0)
Calcium-channel blockers	8,760 (69)	6,800 (68)	1,320 (72)	380 (79)	260 (84)
Thiazide diuretics	3,120 (25)	2,450 (24)	480 (26)	120 (25)	70 (23)
Loop diuretics	1,640 (13)	1,360 (14)	190 (10)	50 (10)	50 (16)
Potassium-sparing diuretics	1,640 (13)	1,270 (13)	230 (13)	80 (17)	60 (19)
Mineralocorticoid receptor antagonists	1,520 (12)	1,180 (12)	210 (11)	70 (15)	60 (19)
Beta-blockers	4,400 (35)	3,220 (32)	760 (41)	250 (52)	180 (58)
Alpha-blockers	1,570 (12)	1,170 (12)	250 (14)	90 (19)	60 (19)
Metformin	1,470 (12)	1,170 (12)	210 (11)	60 (13)	40 (13)
SGLT2i	420 (3)	340 (3)	60 (3)	20 (4)	10 (3)
GLP-1RA	600 (5)	470 (5)	80 (4)	20 (4)	20 (6)
Statins	4,490 (35)	3,610 (36)	600 (33)	160 (33)	120 (39)
Antiplatelets	2,800 (22)	2,260 (23)	380 (21)	90 (19)	80 (26)
Antithrombotic agents	1,030 (8)	800 (8)	150 (8)	50 (10)	30 (10)
Other laboratory measurements					
eGFR levels mL/min/1.73m ²	86.7 (71.0-98.3)	86.2 (70.1-98.0)	87.8 (73.5-99.2)	89.9 (76.8-99.9)	87.7 (71.1-100.8)
Missing	7%	7%	8%	6%	3%
Sodium mmol/L	140.0 (138.5-142.0)	140.0 (138.5-141.5)	141.0 (139.0-142.0)	141.0 (139.0-142.0)	141.0 (140.0-143.0)
Missing	0%	0%	0%	0%	0%
Potassium mmol/L	3.7 (3.4-4.0)	3.8 (3.5-4.0)	3.6 (3.3-3.9)	3.5 (3.3-3.8)	3.4 (3.2-3.6)
Missing	1%	1%	1%	2%	0%
Urine albumin-creatinine ratio mg/mmol	22.0 (9.0-79.0)	22.0 (9.0-80.0)	24.0 (10.0-71.5)	23.0 (10.4-82.0)	39.0 (14.0-113.0)
Missing	44%	45%	42%	40%	39%
Total cholesterol, mmol/L	4.7 (4.1-5.5)	4.8 (4.1-5.5)	4.7 (4.0-5.4)	4.6 (3.9-5.4)	4.6 (4.0-5.1)
Missing	15%	15%	15%	17%	16%
ALT U/L	25.5 (19.0-36.0)	26.0 (19.0-36.0)	25.0 (19.0-35.0)	25.5 (18.0-38.0)	25.0 (18.5-32.5)
Missing	6%	6%	6%	6%	6%

Values are median (Q1-Q3) or n (%), unless otherwise indicated. To protect patient privacy, all numbers were rounded to the nearest 10. ARR 27.7 pmol/mIU, 138.7 pmol/mIU, and 225.8 pmol/mIU are equivalent to 1 ng·dL⁻¹/mIU·L⁻¹, 5 ng·dL⁻¹/mIU·L⁻¹, and 8.1 ng·dL⁻¹/mIU·L⁻¹, respectively. Hypertension is defined by a prescription of ≥2 hypertensive drug classes or a hospital diagnosis of hypertension.

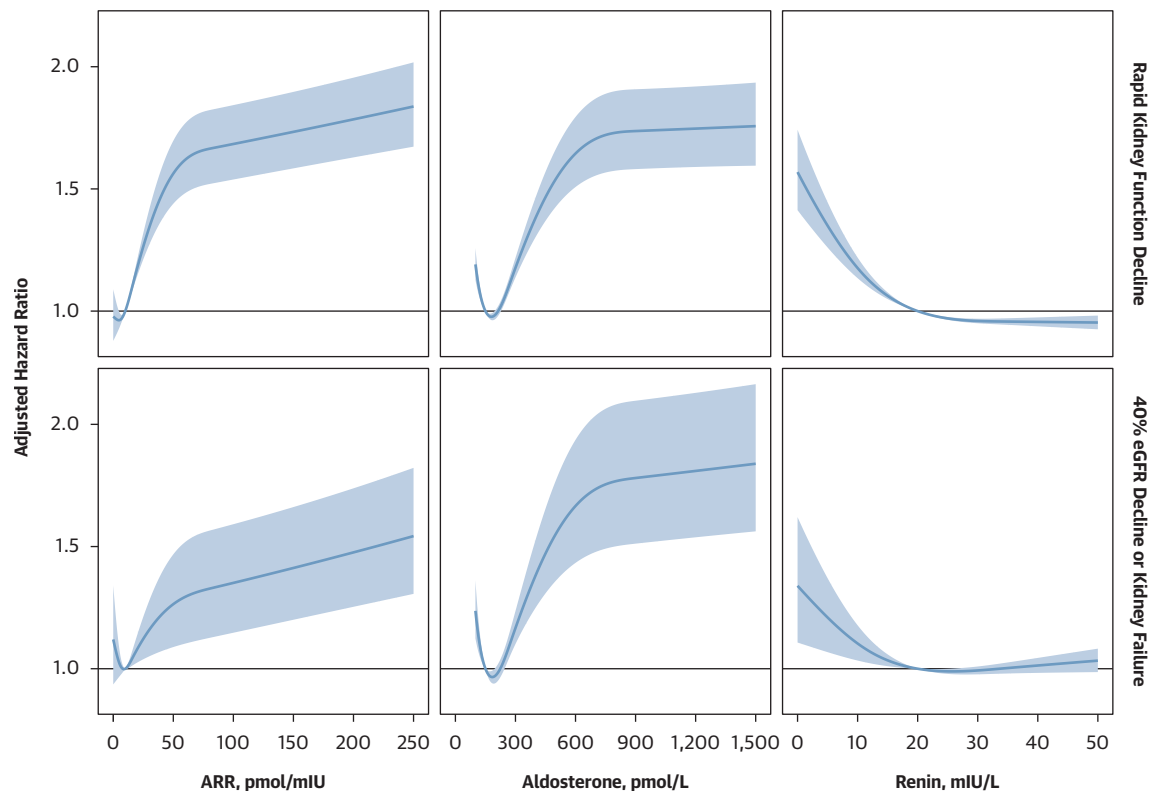
ACE = angiotensin-converting enzyme; ALT = alanine transaminase; ARB = angiotensin II receptor blocker; ARNi = angiotensin II receptor blocker-neprilysin inhibitor; CKD = chronic kidney disease; eGFR = estimated glomerular filtration rate; GLP-1RA = glucagon-like-peptide 1 receptor agonist; SGLT2i = sodium glucose cotransporter 2 inhibitor.

drugs used (0-2 and ≥3). Second, we evaluated additional individual outcomes, including kidney failure, cerebrovascular disease, unstable angina, atrial fibrillation, heart failure hospitalization, peripheral artery disease, and cardiovascular death. Third, to minimize potential protopathic bias where deteriorating cardiovascular or kidney health leading to aldosterone and renin testing, we repeated the analyses excluding cardiorenal events occurring within the first 6 months (180 days) after testing. Fourth, we repeated analyses excluding individuals using mineralocorticoid receptor antagonists and potassium-sparing diuretics, to avoid potential influence of those drugs affecting the ARR results. Fifth, we repeated the analysis excluding all

individuals with primary aldosteronism defined as ARR ≥70 pmol/mIU.³² Finally, we examined the distribution of aldosterone, renin, and ARR and visualized sparse and dense regions in the aldosterone-renin plane.

RESULTS

BASELINE CHARACTERISTICS. We included 12,650 adults with hypertension who underwent ARR testing from 2017 to 2024. The tested population had a median follow-up of 3.6 years (Q1-Q3: 1.9-5.6 years). Of those tested, 10,020 (79%) had an ARR <27.7 pmol/mIU (<1 ng·dL⁻¹/mIU·L⁻¹), 1,840 (15%) 27.7 to <70 pmol/mIU (1 to <2.5 ng·dL⁻¹/mIU·L⁻¹), 480 (4%) 70 to

FIGURE 1 Adjusted Hazard Ratios of Kidney Disease as Restricted Cubic Spline

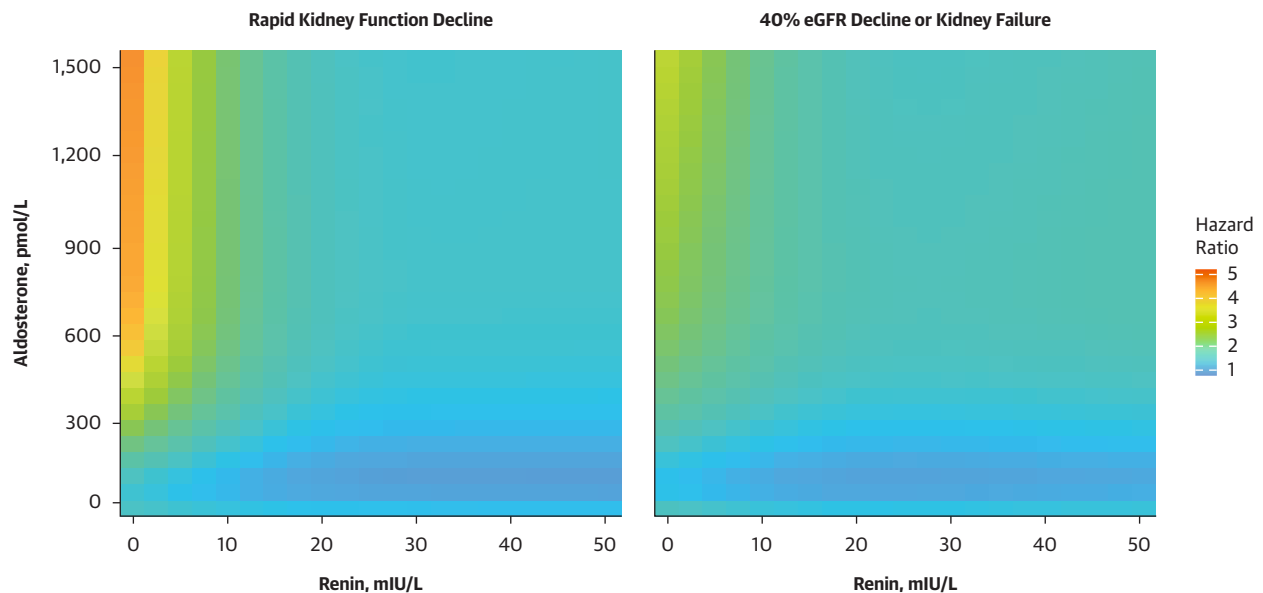
Adjusted HRs (solid lines) and 95% CIs (shaded areas) from restricted cubic spline models of aldosterone-to-renin ratio (ARR), aldosterone, and renin. (Top) Rapid kidney decline (defined by an estimated glomerular filtration rate (eGFR) measurement during follow-up that is ≥ 5 mL/min/1.73 m² lower than the eGFR at inclusion). (Bottom) 40% eGFR decline or kidney failure.

<138.7 pmol/mIU (2.5 to <5 ng·dL⁻¹/mIU⁻¹), and 310 (2%) ≥ 225.8 pmol/mIU (≥ 5 ng·dL⁻¹/mIU⁻¹) (Table 1). Among individuals with ARR <27.7 pmol/mIU, the median aldosterone level was 144 pmol/L (Q1-Q3: 103-308 pmol/L) and the median renin level 28 mIU/L (Q1-Q3: 13-65 mIU/L). In the groups with progressively higher ARR values, corresponding aldosterone concentrations increased substantially, and renin concentrations declined with increasing ARR (Table 1).

The median age was 58 years, and adults with higher ARR levels indicative of renin-independent aldosterone production had a median age similar to tested adults with lower ARR. Comorbidities profiles were similar, including ischemic heart disease, atherosclerotic disease, and dyslipidemia. Median eGFR levels at index date also were similar across ARR groups (87-90 mL/min/1.73 m² in individuals with highest ARR levels vs 86 mL/min/1.73 m² in those with lowest ARR levels). However, those with higher ARR levels had a longer duration of

hypertension (6.7 to 11.2 years vs 6.0 years) and a more intensive antihypertensive therapy (45%-54% vs 39% using ≥ 3 antihypertensive drug classes), with greater use of medications, including calcium channel blocker (72%-84% vs 68%), beta-blockers (41%-58% vs 32%), and potassium-sparing diuretics (13%-19% vs 13%). Despite a greater use of potassium-sparing diuretics, mineralocorticoid receptor antagonists, and angiotensin-converting enzyme inhibitors/angiotensin II receptor blockers, those with higher ARR levels also had lower serum potassium levels (3.4-3.6 mmol/L vs 3.8 mmol/L) and higher urine albumin-creatinine ratios (23-40 mg/mmol vs 22 mg/mmol) (Table 1, Supplemental Table 3).

PRIMARY OUTCOMES. The adjusted HRs (aHRs) of rapid kidney function decline were increased progressively with higher levels of ARR and aldosterone and with lower levels of renin (Figure 1). Consistent with these patterns, the heatmap demonstrated the

FIGURE 2 Heatmap of Adjusted Hazard Ratios of Kidney Disease

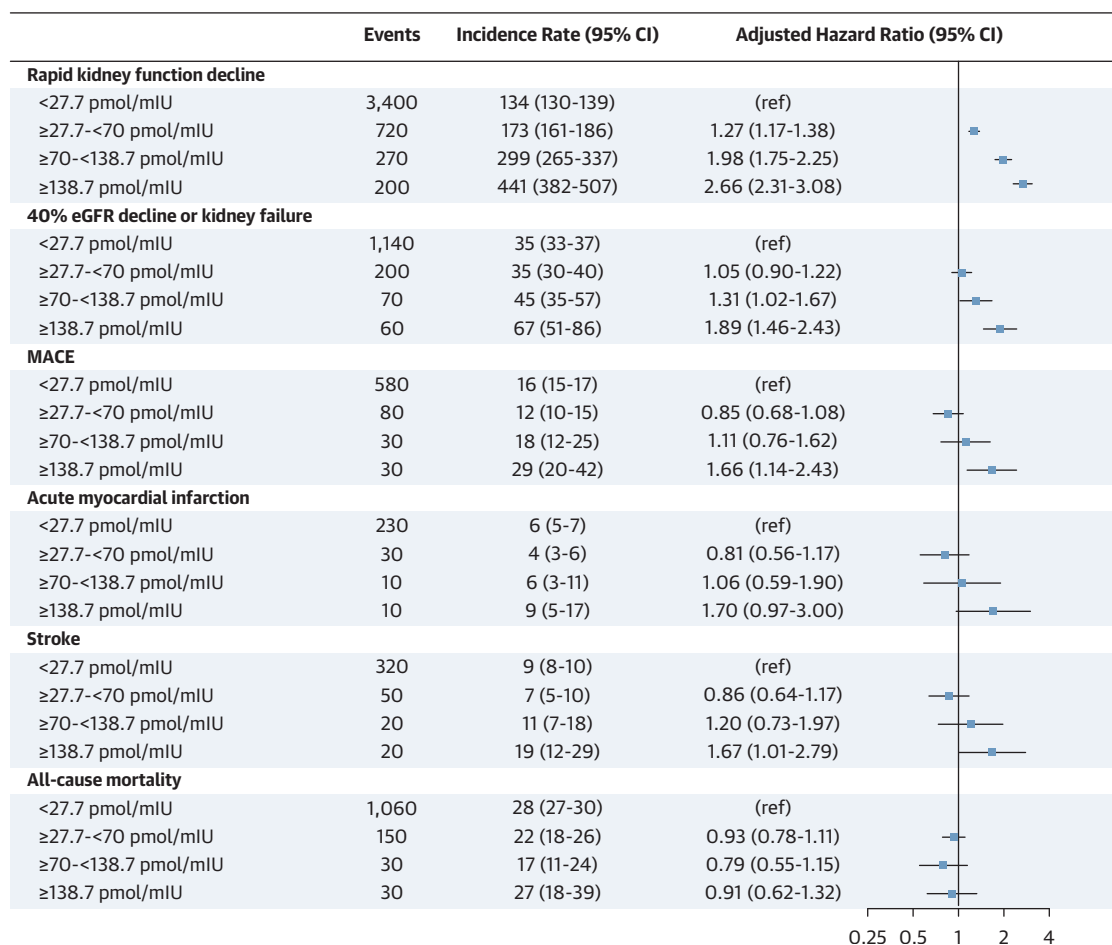
(Left) Rapid kidney decline (defined by an estimated glomerular filtration rate [eGFR] measurement during follow-up that is ≥ 5 mL/min/1.73 m² lower than the eGFR at inclusion). (Right) 40% eGFR decline or kidney failure. Estimates from the map with 95% CIs are provided in [Supplemental Table 2](#).

highest aHR at combinations of high aldosterone and low renin levels with aHR above 2 ([Figure 2](#), [Supplemental Table 4](#)). These associations remained evident also after exclusion of individuals with $\text{ARR} \geq 70$ pmol/mIU in the additional analyses. Analyses using predefined levels for ARR, aldosterone, and renin showed similar trends ([Figures 3-5](#)). Specifically, the aHRs for rapid kidney function decline were 1.27 (95% CI: 1.17-1.38), 1.98 (95% CI: 1.75-2.25), and 2.66 (95% CI: 2.31-3.08) for $\text{ARR} \geq 27.7$ to <70 , ≥ 70 to 138.7, and ≥ 138.7 pmol/mIU, respectively ([Figure 3](#)).

For the composite endpoint of kidney failure or $\geq 40\%$ eGFR decline, aHRs likewise increased with higher ARR and aldosterone levels and lower renin levels ([Figure 1](#)). In line with these patterns, the heatmap demonstrated the highest aHRs at combinations of elevated aldosterone and low renin ([Figure 2](#), [Supplemental Table 4](#)). Analyses using predefined levels for ARR, aldosterone, and renin showed similar trends ([Figures 3-5](#)). For ARR, no increased risk was observed in the lowest level ≥ 27.7 to <70 pmol/mIU (HR: 1.05 [95% CI: 0.90-1.22]), whereas aHRs rose at higher ARR levels: 1.31 (95% CI: 1.02-1.67) and 1.89 (95% CI: 1.46-2.43) for $\text{ARR} \geq 70$ to 138.7 and ≥ 138.7 pmol/mIU, respectively ([Figure 3](#)).

SECONDARY OUTCOMES. For MACE, acute myocardial infarction and stroke, the associations were less pronounced; aHR increased gradually with higher levels of ARR and aldosterone, and had a U-shaped association with renin, with increasing aHR at both lower and higher levels of renin ([Figure 6](#)). Analyses using predefined levels for ARR, aldosterone, and renin showed similar trends ([Figures 3-5](#)). All-cause mortality did not increase with higher ARR, but increased with higher levels of aldosterone and renin ([Figure 6](#)) corresponding with the results using predefined levels ([Figures 3-5](#)). The aHRs for MACE were 0.85 (95% CI: 0.68-1.08), 1.11 (95% CI: 0.76-1.62), and 1.66 (95% CI: 1.14-2.43) for $\text{ARR} \geq 27.7$ to <70 , ≥ 70 to 138.7, and ≥ 138.7 , respectively. For AMI, the corresponding aHRs were 0.81 (95% CI: 0.56-1.17), 1.06 (95% CI: 0.59-1.90), and 1.70 (95% CI: 0.97-3.00), and for stroke they were 0.86 (95% CI: 0.64-1.17), 1.20 (95% CI: 0.73-1.97), and 1.67 (95% CI: 1.01-2.79), respectively.

ADDITIONAL ANALYSES. In subgroup analyses across strata of atherosclerotic cardiovascular disease, heart failure, and chronic kidney disease, associations of ARR, aldosterone, and renin with outcomes were generally consistent with overall results, although estimates were less precise in groups

FIGURE 3 Kidney Disease, Cardiovascular Disease, and Mortality According to ARRs

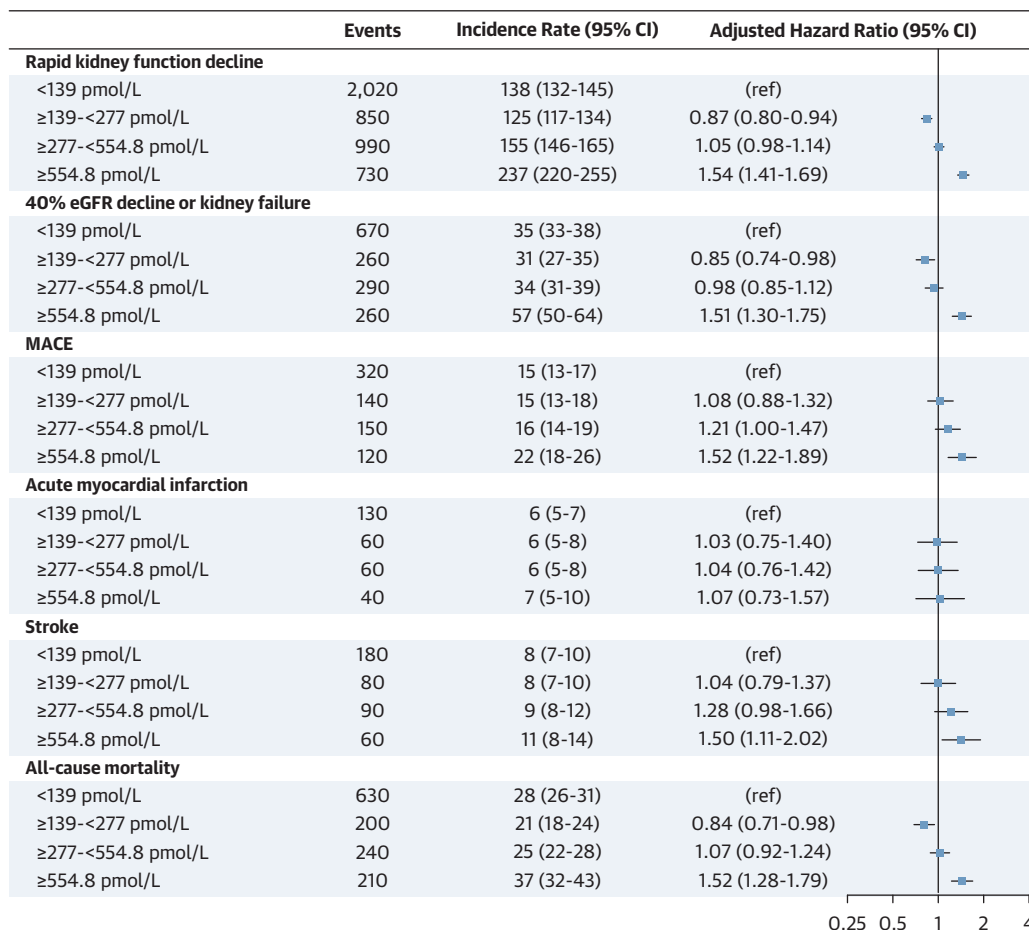
To protect patient privacy, all events were rounded to the nearest 10, and incidence rates were calculated based on these rounded values. ARRs were defined as <27.7, 27.7 to <70.0, 70.0 to <138.7, and ≥138.7 pmol/mIU. 27.7, 70.0, and 138.7 pmol/mIU correspond to 1.0, 2.5, and 5.0 ng·dL⁻¹/mIU·L⁻¹, respectively. Rapid kidney decline is defined by an estimated glomerular filtration rate (eGFR) measurement during follow-up that is ≥5 mL/min/1.73 m² lower than the eGFR at inclusion. ARR = aldosterone-to-renin ratio; MACE = major adverse cardiovascular event.

with fewer events (Supplemental Figures 2-31). Overall, higher levels of ARR and aldosterone were associated with a gradual increase in aHRs, whereas renin showed a U-shaped association with both rapid kidney function decline, the composite of kidney failure or ≥40% eGFR decline, and MACE. These associations were most pronounced among individuals without preexisting comorbidities. In analyses stratified by potassium levels and antihypertensive drug use, associations were also in line with the overall results. Although precision was lower in strata with fewer events, the associations were directionally consistent (Supplemental Figures 32-43).

For the additional outcomes, which included kidney failure, cerebrovascular disease, unstable angina,

atrial fibrillation, heart failure hospitalization, peripheral artery disease, and cardiovascular death, we found no consistent associations with ARR, aldosterone, or renin (Supplemental Figures 44-47). Event numbers were low, limiting precision of level and spline analyses. For kidney failure, aHR for ARR ≥27.7 to <70, ≥70 to 138.7, and ≥138.7 pmol/mIU were 1.18 (95% CI: 0.81-1.71), 0.66 (95% CI: 0.27-1.62), and 0.68 (95% CI: 0.25-1.85), respectively. For atrial fibrillation, the corresponding estimates were 1.05 (95% CI: 0.77-1.43), 1.14 (95% CI: 0.66-1.96), and 1.65 (95% CI: 0.93-2.90).

In analyses excluding events occurring within the first 6 months after testing, associations of ARR, aldosterone, and renin with outcomes were generally

FIGURE 4 Kidney Disease, Cardiovascular Disease, and Mortality According to Aldosterone Levels

To protect patient privacy, all events were rounded to the nearest 10, and incidence rates were calculated based on these rounded values. Aldosterone levels were defined as <139, 139 to <277, 277 to <554.8, and ≥554.8 pmol/L. Rapid kidney decline is defined by an estimated glomerular filtration rate (eGFR) measurement during follow-up that is ≥5 mL/min/1.73 m² lower than the eGFR at inclusion. MACE = major adverse cardiovascular event.

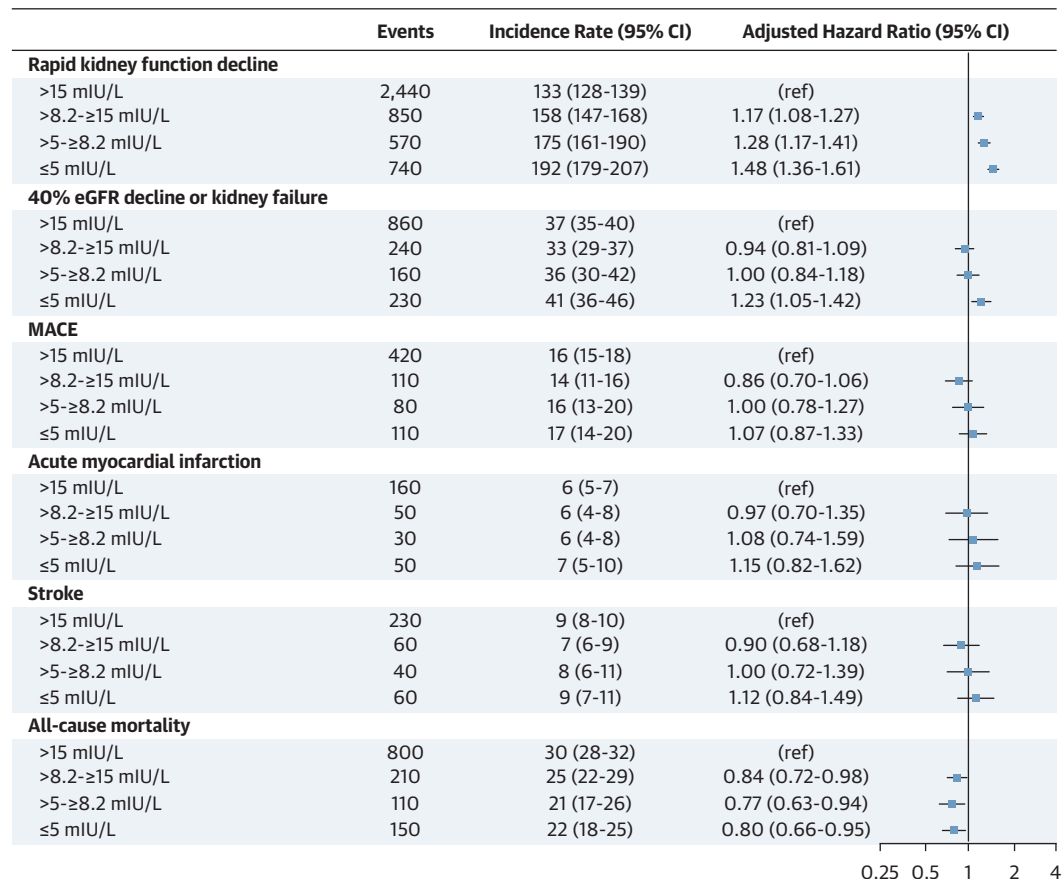
consistent with the main findings (Supplemental Figures 48-50). For rapid kidney function decline, the aHRs for ARR 27.7 to <70, 70 to <138.7, and ≥138.7 pmol/mIU were, respectively, 1.16 (95% CI: 1.05-1.28), 1.85 (95% CI: 1.59-2.15), and 2.24 (95% CI: 1.88-2.67). For ≥40% decline in eGFR or kidney failure, the respective estimates were 0.99 (95% CI: 0.85-1.16), 1.26 (95% CI: 0.98-1.62), and 1.95 (95% CI: 1.51-2.51) (Supplemental Figures 51-53).

In analyses excluding individuals using potassium-sparing diuretics or mineralocorticoid receptor antagonists, results were overall similar to the main findings (Supplemental Figures 54-58). Higher levels of ARR and aldosterone were associated with a gradual increase in aHRs, whereas renin

demonstrated U-shaped associations with rapid kidney function decline, the composite of kidney failure or ≥40% eGFR decline, and MACE.

In analyses excluding individuals with ARR ≥70 pmol/mIU, rapid kidney function decline remained clearly associated with higher ARR and aldosterone levels and lower renin levels. Associations for the other outcomes showed a similar tendency, although they were less consistent and pronounced (Supplemental Figures 59-61).

Renin concentrations were predominantly low, with a median of 20 mIU/L and mostly <49 mIU/L. Aldosterone concentrations were likewise concentrated at the lower end, with a median of 178 pmol/L and mostly <370 pmol/L. The heatmap with bivariate

FIGURE 5 Kidney Disease, Cardiovascular Disease, and Mortality According to Renin Levels

To protect patient privacy, all events were rounded to the nearest 10, and incidence rates were calculated based on these rounded values. Renin levels were defined as >15, >8.2 to 15, >5.0 to 8.2, and ≤5.0 mIU/L. Rapid kidney decline is defined by an estimated glomerular filtration rate (eGFR) measurement during follow-up that is ≥5 mL/min/1.73 m² lower than the eGFR at inclusion. MACE = major adverse cardiovascular event.

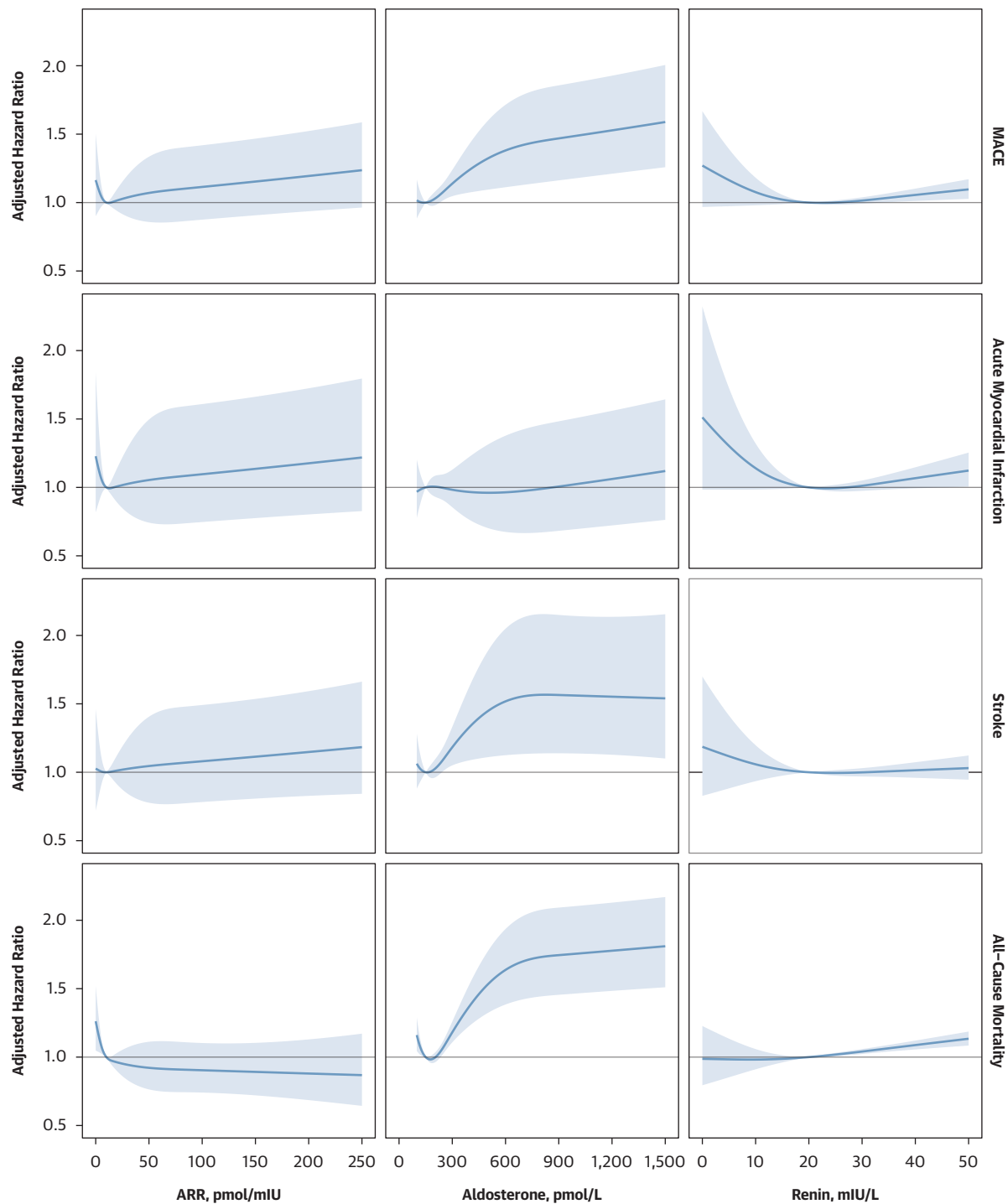
distribution showed the highest density in the region of low aldosterone and low renin, with only sparse observations at higher concentrations of either (Figure 7, Supplemental Figure 62).

DISCUSSION

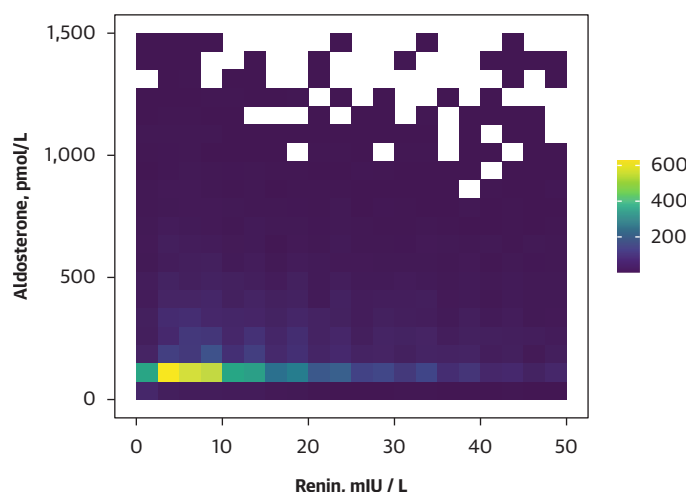
In this large nationwide population-based cohort study of 12,230 individuals with hypertension undergoing routine screening, we found that renin-independent aldosterone production, across a spectrum of ARR and aldosterone and renin levels, was associated with increased risks of renal function decline and MACE. To our knowledge, this is the first study demonstrating these associations in a nationwide study with complete laboratory data. Our

findings were generally consistent across a range of subgroup analyses, and across continuous measures of ARR, aldosterone, and renin with the use of splines and heatmaps and with predefined levels in forest plots.

Our results extend the previous evidence linking renin-independent aldosterone production to adverse cardiorenal outcomes in tested populations. In CONPASS (n = 2,556 individuals with hypertension), the OR of cardiovascular disease increased from 1.02 (95% CI: 0.62-1.86) at an ARR ≥27.7 pmol/mIU (≥1 ng·dL⁻¹/mIU·L⁻¹), to 1.68 (95% CI: 0.82-3.43) at ARR ≥136 pmol/mIU (≥4.9 ng·dL⁻¹/mIU·L⁻¹).¹⁷ Similarly, in FOS (n = 1,433 participants with hypertension), the HRs plateaued around ARR ≥27.7 pmol/mIU (≥1 ng·dL⁻¹/mIU·L⁻¹) with a value of 1.49 (95% CI:

FIGURE 6 Adjusted Hazard Ratios of Cardiovascular Disease as Restricted Cubic Spline

Adjusted HRs (solid lines) and 95% CIs (shaded areas) from restricted cubic spline models of aldosterone-to-renin ratio (ARR), aldosterone, and renin for major adverse cardiovascular events (MACE), acute myocardial infarction, stroke, and all-cause mortality.

FIGURE 7 Bivariate Density Heatmap of Aldosterone and Renin Measurements

Bivariate distribution of plasma aldosterone and renin measurements among adults with hypertension undergoing clinically indicated testing.

1.19-1.86).¹⁷ The Jackson Heart Study reported an HR of 1.31 (95% CI: 1.02-1.68) for incident cardiovascular disease among participants in the upper median of aldosterone levels compared with the lower median.⁶ In ARIC (n = 5,980 participants), each doubling of ARR was associated with an HR of 1.04 (95% CI: 1.01-1.08) for a composite cardiovascular outcome.¹⁸ Most recently, the CARTaGENE study of 2,017 relatively healthy Canadian adults (of whom only 27% had known hypertension) found that higher ARR and lower renin were associated with increased risk of MACE, independently from blood pressure.¹⁹ In contrast to those studies, the present study examined >12,000 adults with known hypertension across an entire nation and focused specifically on kidney function decline and failure, with additional assessments of cardiovascular and mortality outcomes. Renin displayed a U-shaped association with cardiovascular events. In contrast to low renin, high renin is most likely related to other conditions, eg, heart failure, renin-angiotensin-aldosterone system inhibitor use, and renal artery stenosis.

Evidence linking aldosterone to kidney disease progression has been reported across multiple studies. In a registry-based cohort of 1,334 patients undergoing aldosterone testing in routine care in the United States, elevated plasma aldosterone levels (≥ 5 ng/dL, >138.7 pmol/L) were associated with elevated risk of hypertension (OR: 2.01; 95% CI: 1.38-2.92), and higher levels of aldosterone (≥ 10 ng/dL, >277.4 pmol/L) were further associated with chronic

kidney disease (OR: 1.49; 95% CI: 1.15-1.92).³⁷ In another study of 3,680 patients with chronic kidney disease, individuals in the highest quartile of serum aldosterone had an increased risk of chronic kidney disease progression compared with those in the lowest quartile (HR: 1.45; 95% CI: 1.22-1.73).¹⁴ Similarly, among patients with hypertension and metabolic disease, the highest plasma aldosterone tertile was associated with increased risk of renal function decline compared with the lowest tertile (HR: 1.87; 95% CI: 1.09-3.19).³⁸ Taken together, these findings and our results highlight renin-independent aldosterone production as a clinically relevant risk factor for adverse cardiovascular and renal outcomes. This study supports a paradigm shift in perspective, from viewing primary aldosteronism physiology as a rare, binary condition to recognizing it as a continuum that may contribute to adverse outcomes.¹⁷

A total of 780 individuals (6.4%) in our routine care cohort of adults with hypertension met Endocrine Society criteria for primary aldosteronism (ARR ≥ 70 pmol/mIU).³¹ These individuals were in the highest ARR and aldosterone levels and the lowest renin levels, and their presence likely contributes to the steep risk gradients observed at the upper end of the dose-response spectrum. Importantly, the overall associations with rapid kidney function decline persisted after excluding these individuals, demonstrating that the risk for incident kidney disease parallels the full continuum of renin-independent aldosterone production, transcending the currently used definitions for primary aldosteronism. In this manner, our findings indicate that primary aldosteronism is a syndrome of renin-independent aldosterone production, wherein aldosterone produced when renin is low appears to be associated with risk for incident kidney disease in a dose-dependent manner.

Several pathophysiologic mechanisms may explain the observed associations. By activating the mineralocorticoid receptor, aldosterone exerts direct adverse effects on cardiovascular and renal tissues that go beyond blood pressure regulation.⁹ Experimental and clinical studies have implicated the activation of mineralocorticoid receptor by aldosterone in renal fibrosis, glomerular hypertrophy, vascular inflammation, oxidative stress, and endothelial dysfunction, all of which contribute to progressive kidney damage as well as the development of atherosclerosis and cardiovascular damage.^{9,15,16}

A major strength of our study is its nationwide population-based design, which enabled linkage of complete nationwide laboratory data and near-complete follow-up for outcomes.²⁵ Importantly,

CENTRAL ILLUSTRATION Renin-Aldosterone Profiles and Cardiorenal Outcomes in Hypertension: A Nationwide Cohort Study

ADEL Study: Population and Research Question



Danish population-based cohort study (2017-2024)



12,650 adults (≥18 years) with HTN undergoing routine aldosterone and renin testing



Median age: 58 years



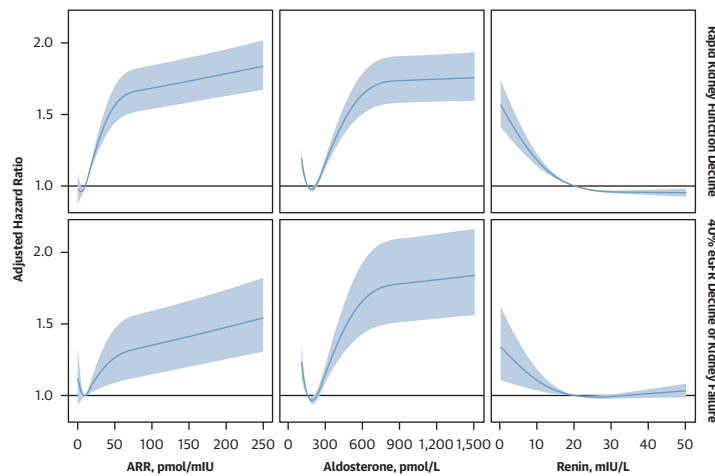
Largest sample size of its kind to date

Data sources:
Four Danish National Registries^a

Median follow-up: 3.6 years (IQR 1.9-5.6 years)

What is the impact of aldosterone dysregulation on cardiorenal outcomes in adults with hypertension?

Primary Outcomes: Kidney Function



Risk of rapid kidney function decline^b and
≥40% eGFR decline or kidney failure increased progressively with:

↑ levels of aldosterone and aldosterone-to-renin ratio (ARR)

and

↓ levels of renin

CV Outcomes

Lower ARR levels (≥27.7-70 pmol/mIU)



Risk for MACE not increased (aHR 0.86)

Higher ARR thresholds (≥70-138.7 and ≥138.7 pmol/mIU)



Increased risk for MACE (aHR 1.20 and 1.67)

- Among individuals with HTN, higher ARR, higher aldosterone, and lower renin were associated with increased risk of kidney function decline
- Higher ARR thresholds were associated with greater risk of MACE
- This underscores the prognostic importance of aldosterone dysregulation in adults with HTN

Ljungberg C, et al. JACC. 2026;■(■):■-■.

^aDanish Civil Registration System, Danish National Patient Registry, Danish National Prescription Registry, and Registry of Laboratory Results for Research. ^bRapid kidney function decline was defined as an annual eGFR decrease of ≥5 mL/min/1.73 m² per year. ADEL = Aldosterone Dysregulation Epidemiology Landscape; ARR = aldosterone-to-renin-ratio; CV = cardiovascular; eGFR = estimated glomerular filtration rate; HTN = hypertension; MACE = major adverse cardiovascular events.

our findings reflect the real-world population of patients selected for ARR testing, which strengthens the clinical relevance albeit limiting applicability to the broader hypertensive population. This allowed us to include all individuals tested for aldosterone and renin across both primary and secondary care settings, resulting in a large sample size. With the use of splines and multiple ARR categories, ranging from guideline-based cutoffs to empirically derived values, we were able to characterize risk gradients across the full range of ARR-defined aldosterone-renin profiles. Nonetheless, this study has limitations. First, the study population was restricted to individuals who underwent ARR testing based on clinical judgement and criteria. The Danish clinical guidelines during our study period recommended ARR testing in a highly selected subset of patients, ie, preferably patients with difficult-to-treat hypertension who needed 3 or more antihypertensive drugs or had other indications of possible primary aldosteronism, such as low potassium levels.³⁹ Thus, our study population likely represents a higher-risk subset of hypertensive patients rather than the general hypertensive population. As such, the ARR-tested group is likely to differ systematically from the broader hypertensive population in terms of comorbidities and clinical characteristics, limiting generalizability. In addition, for those tested, the indication for testing may reflect other, unmeasured clinical concerns, further complicating interpretation. Second, although we used a validated algorithm to define hypertension,³⁰ some misclassification remains possible. Nevertheless, similar associations with cardiorenal outcomes were observed in the subgroup analyses not requiring hypertension, supporting the robustness of our findings. Third, ARR test results may be affected by several preanalytical factors, including physical activity, time of day, salt intake, diet, and medication use.⁴⁰ Our data could not capture whether medications were temporarily discontinued before ARR testing, but the additional analyses excluding individuals using potassium-sparing diuretics or mineralocorticoid receptor antagonists were similar to the main findings. ARR was used as it is considered less susceptible of intra- and inter-individual variability.⁴¹⁻⁴³ Nonetheless, the typical absence of confirmatory ARR testing in our routine care data implies that the occurrence of renin-independent aldosterone production in our study may have been under- or overestimated. Any resulting misclassification is expected to be non-differential in terms of clinical outcome and would likely bias associations toward the null. We examined the variability of ARR within individuals with a

second test (2,900 individuals) and found modest within-person variability ([Supplemental Table 5](#)). Fourth, we lacked information on blood pressure, body mass index, socioeconomic factors, and lifestyle behaviors. Despite adjustment for a wide range of variables, residual confounding and selection bias cannot be excluded. Finally, as in other published studies, our data had a relatively short follow-up period in the context of hypertension and therefore low event rates for some subgroups, reflected by possibly imprecise point estimates with wide confidence intervals. Nevertheless, this study has the largest sample size to date compared with others.

CONCLUSIONS

Renin-aldosterone physiology and pathophysiology represents a continuous risk gradient, not a binary abnormality, transcending arbitrary thresholds. Kidney outcomes appear earlier and more consistently than cardiovascular outcomes. Even below classic thresholds for primary aldosteronism, renin-independent aldosterone production may identify patients at increased risk of progressive kidney disease in a dose-dependent manner ([Central Illustration](#)). Finally, ARR-tested patients represent a high-risk clinical subset; results should be interpreted in this context.

DATA AVAILABILITY The study was registered at the Danish Data Protection Agency through registration at Aarhus University. Under Danish law, researchers are not permitted to share raw registry data with third parties. To protect patient privacy, the combined data set used in this study is accessible only through the National Danish Health Data Authority.

FUNDING SUPPORT AND AUTHOR DISCLOSURES

This study was funded by a grant from AstraZeneca to Aarhus University. The Department of Clinical Epidemiology is involved in other studies with funding from various companies as research grants to, and administered by, Aarhus University. Dr Thomsen has given occasional single presentations on medical research (with or without financial compensation to self or Aarhus University, no permanent affiliation to any company) for companies including AstraZeneca, Bayer, Boehringer Ingelheim, Eli Lilly, Novo Nordisk, and Sanofi. Dr Christiansen has served, on behalf of Aarhus University, as a scientific consultant for AstraZeneca on the *Inside CKD* study (with financial compensation to Aarhus University). Dr Bech has participated in advisory boards organized by AstraZeneca and Bayer. Drs Carlsson, Shojaiyan, Franzén, and Chin are employees of and own stock in AstraZeneca. Dr Ljungberg has reported that she has no relationships relevant to the contents of this paper to disclose.

ADDRESS FOR CORRESPONDENCE: Dr Christine Ljungberg, Olof Palmes Allé 43-45, 8200 Aarhus N, Denmark. E-mail: chrlju@clin.au.dk.

REFERENCES

1. Lifton RP, Gharavi AG, Geller DS. Molecular mechanisms of human hypertension. *Cell*. 2001;104:545-556.
2. Vecchiola A, Lagos CF, Carvajal CA, Baudrand R, Fardella CE. Aldosterone production and signaling dysregulation in obesity. *Curr Hypertens Rep*. 2016;18:20.
3. Vaidya A, Underwood PC, Hopkins PN, et al. Abnormal aldosterone physiology and cardiometabolic risk factors. *Hypertension*. 2013;61:886-893.
4. Funder JW, Reincke M. Aldosterone: a cardiovascular risk factor? *Biochim Biophys Acta*. 2010;1802:1188-1192.
5. Parksook WW, Williams GH. Aldosterone and cardiovascular diseases. *Cardiovasc Res*. 2023;119:28-44.
6. Joseph JJ, Echouffo-Tcheugui JB, Kalyani RR, et al. Aldosterone, renin, cardiovascular events, and all-cause mortality among African Americans: the Jackson Heart Study. *JACC Heart Fail*. 2017;5:642-651.
7. Brown JM, Robinson-Cohen C, Luque-Fernandez MA, et al. The spectrum of subclinical primary aldosteronism and incident hypertension: a cohort study. *Ann Intern Med*. 2017;167:630-641.
8. Brown JM, Siddiqui M, Calhoun DA, et al. The unrecognized prevalence of primary aldosteronism: a cross-sectional study. *Ann Intern Med*. 2020;173:10-20.
9. Brown JM. Adverse effects of aldosterone: beyond blood pressure. *J Am Heart Assoc*. 2024;13:e030142.
10. Vecchiola A, Fuentes CA, Barros ER, et al. The aldosterone/renin ratio predicts cardiometabolic disorders in subjects without classic primary aldosteronism. *Am J Hypertens*. 2019;32:468-475.
11. Hundemer GL, Agharazii M, Madore F, et al. Subclinical primary aldosteronism and cardiovascular health: a population-based cohort study. *Circulation*. 2024;149:124-134.
12. Buffolo F, Tetti M, Mulatero P, Monticone S. Aldosterone as a mediator of cardiovascular damage. *Hypertension*. 2022;79:1899-1911.
13. Polenus P, Đuran A, Karanović Štambuk S. Aldosterone in chronic kidney disease. *Bio-medicines*. 2025;13:2081.
14. Verma A, Vaidya A, Subudhi S, Waikar SS. Aldosterone in chronic kidney disease and renal outcomes. *Eur Heart J*. 2022;43:3781-3791.
15. Brown JM, Wijkman MO, Claggett BL, et al. Cardiac structure and function across the spectrum of aldosteronism: the Atherosclerosis Risk In Communities study. *Hypertension*. 2022;79:1984-1993.
16. Hu J, Shen H, Huo P, et al. Heightened cardiovascular risk in hypertension associated with renin-independent aldosteronism versus renin-dependent aldosteronism: a collaborative study. *J Am Heart Assoc*. 2021;10:e023082.
17. He C, Li R, Yang J, et al. Optimizing the aldosterone-to-renin ratio cutoff for screening primary aldosteronism based on cardiovascular risk: a collaborative study. *Clin Exp Hypertens*. 2024;46:2301571.
18. Lassen M, Ostrominski J, Claggett B, et al. Aldosterone dysregulation and risk of cardiovascular outcomes: the Atherosclerosis Risk In Communities study. *J Am Coll Cardiol*. 2025;85, 665-665.
19. Goupil R, Desbiens LC, Merabtine A, et al. Subclinical primary aldosteronism and major adverse cardiovascular events: a longitudinal population-based cohort study. *Circulation*. 2025;152:913-923.
20. Inoue K, Goldwater D, Allison M, Seeman T, Kestenbaum BR, Watson KE. Serum aldosterone concentration, blood pressure, and coronary artery calcium: the Multi-Ethnic Study of Atherosclerosis. *Hypertension*. 2020;76:113-120.
21. Newton-Cheh C, Guo CY, Gona P, et al. Clinical and genetic correlates of aldosterone-to-renin ratio and relations to blood pressure in a community sample. *Hypertension*. 2007;49:846-856.
22. Vasan RS, Evans JC, Larson MG, et al. Serum aldosterone and the incidence of hypertension in nonhypertensive persons. *N Engl J Med*. 2004;351:33-41.
23. Helmecci W, Hundemer GL. Targeting aldosterone to improve cardiorenal outcomes: from nonsteroidal mineralocorticoid receptor antagonists to aldosterone synthase inhibitors. *Curr Opin Nephrol Hypertens*. 2025.
24. Ljungberg C, Heide-Jørgensen U, Carlsson B, et al. Occurrence and patient characteristics of aldosterone dysregulation in routine clinical care: a population-based study. *JACC Adv*. 2025;5:102385.
25. Schmidt M, Schmidt SAJ, Adelborg K, et al. The Danish health care system and epidemiological research: from health care contacts to database records. *Clin Epidemiol*. 2019;11:563-591.
26. Schmidt M, Pedersen L, Sørensen HT. The Danish Civil Registration System as a tool in epidemiology. *Eur J Epidemiol*. 2014;29:541-549.
27. Schmidt M, Schmidt SA, Sandegaard JL, Ehrenstein V, Pedersen L, Sørensen HT. The Danish National Patient Registry: a review of content, data quality, and research potential. *Clin Epidemiol*. 2015;7:449-490.
28. Pottegård A, Schmidt SAJ, Wallach-Kildemoes H, Sørensen HT, Hallas J, Schmidt M. Data resource profile: the Danish National Prescription Registry. *Int J Epidemiol*. 2017;46, 798-798f.
29. Arendt JFH, Hansen AT, Ladefoged SA, Sørensen HT, Pedersen L, Adelborg K. Existing data sources in clinical epidemiology: laboratory information system databases in Denmark. *Clin Epidemiol*. 2020;12:469-475.
30. Bonnesen K, Schmidt M. Validity of prescription-defined and hospital-diagnosed hypertension compared with self-reported hypertension in Denmark. *Clin Epidemiol*. 2024;16:249-256.
31. Adler GK, Stowasser M, Correa RR, et al. Primary aldosteronism: an Endocrine Society clinical practice guideline. *J Clin Endocrinol Metab*. 2025;110:2453-2495.
32. Mulatero P, Monticone S, Deinum J, et al. Genetics, prevalence, screening and confirmation of primary aldosteronism: a position statement and consensus of the Working Group on Endocrine Hypertension of The European Society of Hypertension. *J Hypertens*. 2020;38:1919-1928.
33. Townsend R, Agiro A, Luan S, et al. Aldosterone dysregulation and type 2 diabetes mellitus are associated with the risk of hypertension: BREAKTHROUGH Risk study. *Metab Clin Exp*. 2025;168(Suppl):156203.
34. Townsend R, Agiro A, Luan S, et al. Aldosterone dysregulation is a risk factor for hypertension and chronic kidney disease: BREAKTHROUGH Risk study. *Am J Hypertens*. 2026;39:161-170.
35. Kidney Disease: Improving Global Outcomes (KDIGO) CKD Work Group. KDIGO 2024 clinical practice guideline for the evaluation and management of chronic kidney disease. *Kidney Int*. 2024;105:S117-S314.
36. Eknoyan G, Lameire N, Eckardt K, et al. KDIGO 2012 clinical practice guideline for the evaluation and management of chronic kidney disease. *Kidney Int*. 2013;3:5-14.
37. Townsend RR, Agiro A, Luan S, et al. Risk of hypertension and chronic kidney disease following aldosterone dysregulation. *Am J Hypertens*. 2026;39:161-170.
38. Lin M, Heizhati M, Gan L, et al. Higher aldosterone is associated with increased renal impairment risk in patients with hypertension and abnormal glucose metabolism: a longitudinal study. *J Hypertens*. 2022;40:561-569.
39. Hornung N. *Lægehåndbogen aldosteron*. 2023.
40. Hundemer GL, Agharazii M, Madore F, et al. Sex-specific associations of aldosterone and renin with body composition: a population-based cohort study. *J Clin Endocrinol Metab*. 2024;110:801-810.
41. Mulatero P, Sechi LA, Williams TA, et al. Subtype diagnosis, treatment, complications and outcomes of primary aldosteronism and future direction of research: a position statement and consensus of the Working Group on Endocrine Hypertension of the European Society of Hypertension. *J Hypertens*. 2020;38:1929-1936.
42. Ng E, Gwini SM, Libianto R, et al. Aldosterone, renin, and aldosterone-to-renin ratio variability in screening for primary aldosteronism. *J Clin Endocrinol Metab*. 2022;108:33-41.
43. Veldhuizen GP, Alnazer RM, Kroon AA, de Leeuw PW. Variability of aldosterone, renin and the aldosterone-to-renin ratio in hypertensive patients without primary aldosteronism. *J Hypertens*. 2022;40:2256-2262.

KEY WORDS aldosterone dysregulation, clinical practice, cohort study, epidemiology, hypertension

APPENDIX For supplemental tables and figures, please see the online version of this paper.